

MATERIALS & METHODS — POPULATION GENETICS

Demographic inference of effective population size $N_e(t)$ in *Dicorynia guianensis*

Scientific and methodological description of Stairway Plot 2 and SMC++ analyses, intended for the Materials & Methods section of a scientific article.

155 individuals — French Guiana **4** ADMIXTURE clusters ($K = 4$) ● Pop_1 ● Pop_2 ● Pop_3 ● Pop_4

$\mu = 5 \times 10^{-9}$ | $T_g = 30$ yr

COMMON INPUT

0. Shared input data

Demographic analyses were performed on the **155 French Guiana individuals** of *Dicorynia guianensis*, assigned to the four genetic clusters (Pop_1 to Pop_4) identified by ADMIXTURE at $K = 4$. The four Cameroonian and Beninese outgroup individuals (DBR_AGN, DBR_AGP, DBR_AGQ, DBR_AGR; site Cameroun_Benin) were excluded from all demographic analyses to avoid biasing N_e estimates through inter-continental divergence or ancestral introgression.

The reference mutation rate is $\mu = 5 \times 10^{-9}$ substitutions per site per generation. The generation time is set to **30 years/generation** for conversion of temporal axes from generations to calendar years. A mutation rate sensitivity analysis ($\mu \in \{2, 3, 4, 5, 6, 7, 8\} \times 10^{-9}$) is performed for both methods.

METHOD 1 — SFS-BASED

1. Stairway Plot 2

1.1 Rationale

Stairway Plot 2 (Liu & Fu 2015, 2020; v2.1.2) is a non-parametric demographic inference method that reconstructs temporal variation in effective population size

$N_e(t)$ from the **site frequency spectrum (SFS)**. A stepwise (stairway) model is fitted to the observed SFS by composite-likelihood maximisation, without assuming any specific demographic model a priori. Confidence intervals (2.5–97.5%) are obtained by bootstrap over random subsets of the SFS.

The method uses only the *distribution* of allele frequencies, not the spatial arrangement of variants along the genome. It assumes statistical independence between sites.

1.2 SFS preparation

The input VCF is the **LD-pruned** version of the TE-filtered dataset (`snpS_TE_filtered.ld_pruned.vcf.gz` , step 06.6b). Linkage disequilibrium pruning is applied to respect Stairway Plot 2's site-independence assumption: sites in strong LD violate this assumption and bias the SFS through overrepresentation of certain allele frequencies.

For each cluster, a subset VCF is extracted with bcftools v1.9, retaining only biallelic SNPs polymorphic *within* the cluster ($MAC \geq 1$; option `--min-ac 1:minor`). The folded SFS (minor allele; folded SFS) is computed with easySFS v0.0.1:

- **Projection:** full $2n$ ($2 \times$ number of individuals in the cluster) — no downsampling of observed chromosomes
- **Folded mode** (`--ploidy 2`): ancestral/derived polarisation is not available with certainty
- Option `-a` : all SNPs retained (appropriate for whole-genome sequencing data, not RADseq)
- The SFS is truncated to bins 1 to $n_{seq}/2$; bin 0 (monomorphic sites) excluded; bins $> n_{seq}/2$ are zero by construction of the folded SFS

1.3 Effective genome length (L)

The parameter L represents the number of observable callable sites, used to normalise SFS information and convert $\theta = 4 \cdot N_e \cdot \mu$ to N_e . It was estimated separately for each cluster (step 07.11b) by counting sites with sufficient coverage ($GQ \geq 20$) located outside transposable element regions, from 5 representative individuals per population (individual GVCFs). The mean was used as the L value in the blueprint.

POPULATION	L — CALLABLE SITES (BP)	NOTE
● Pop_1	215,357,483	
● Pop_2	221,273,354	
● Pop_3	193,750,488	Lower value — reduced GQ coverage in this cluster; unrelated to demography
● Pop_4	216,089,077	

Total non-TE genome: 270,668,577 bp.

1.4 Inference parameters

PARAMETER	VALUE	DESCRIPTION / RATIONALE
<code>mu</code>	5×10^{-9}	Reference mutation rate (sub./site/gen.)
<code>year_per_generation</code>	30 yr	Generation time estimated for <i>D. guianensis</i>
<code>ninput</code>	200	Bootstrap replicates
<code>pct_training</code>	0.67	67% of replicates for training, 33% for testing
<code>whether_folded</code>	true	Folded SFS (ancestral polarisation unavailable)
<code>nrand</code> (large pop.)	7, 15, 22, 28	If $nseq/2 \geq 28$: standard values
<code>nrand</code> (small pop.)	HALF/4, HALF/2, 3·HALF/4, HALF	If $nseq/2 < 28$: proportionally scaled; strict monotonicity enforced

The `nrand` parameter specifies the four model-complexity levels (number of temporal segments) explored during optimisation. Proportional scaling for smaller populations prevents overparameterisation (number of breakpoints $\leq nseq/2$).

1.5 Execution

Inference proceeds in three phases:

1. **Stairbuilder** (phase 1): generates the training script from the blueprint.
2. **Replicate training** (phase 2): 200 bootstrap replicates are submitted in parallel via `sarray` (individual SLURM array jobs). Each replicate fits the model on a random

67% subset of the SFS.

3. **Summary** (phase 3): aggregates 200 replicates and computes the median and confidence intervals (2.5%, 12.5%, 87.5%, 97.5%) for $N_e(t)$ at each time point.

The four clusters were processed sequentially (memory constraint when calling easySFS on large SNP sets). The 200 training replicates were submitted in parallel per cluster.

1.6 Mutation rate sensitivity analysis

Uncertainty in μ is a major source of error in converting θ to N_e and absolute time. When μ changes from μ_{ref} to μ_{alt} , both axes (N_e and time) rescale by the factor $\mu_{\text{ref}} / \mu_{\text{alt}}$ (from the property $\theta = 4 \cdot N_e \cdot \mu$). A sensitivity analysis was performed by applying this rescaling for:

$$\mu \in \{2, 3, 4, 5, 6, 7, 8\} \times 10^{-9} \text{ sub./site/gen.}$$

The reference value ($\mu = 5 \times 10^{-9}$) is shown in the main combined figure; alternative values are displayed to quantify the total uncertainty envelope on $N_e(t)$ trajectories.

METHOD 2 — SPATIAL COALESCENT

2. SMC++ — per-population estimation

2.1 Rationale

SMC++ (Terhorst *et al.* 2017; v1.15.5) reconstructs $N_e(t)$ by exploiting the **spatial arrangement of variants along the genome** through a piecewise Sequentially Markov Coalescent (SMC) model. Unlike Stairway Plot, SMC++ uses information encoded in the *distribution of inter-variant distances* — not just the SFS — which provides better temporal resolution for recent demographic changes. SMC++ leverages multiple "distinguished individuals" to multiply independent coalescent signals within a single population.

2.2 Input data

The input VCF is the **TE-filtered but not LD-pruned** dataset

(`snps_TE_filtered.vcf.gz` , step 06.5). LD pruning is not applied because SMC++

explicitly exploits the spatial information of variants: removing sites in strong LD would destroy the coalescent information encoded in local haplotype structure.

A shared VCF comprising the 155 French Guiana individuals was extracted with bcftools, restricted to the **13 chromosome-level Super-Scaffolds** (≥ 30 Mb). Scaffold names are used directly without integer renaming (unlike GONE/PLINK analyses).

2.3 Transposable element mask

TE regions (`te_merged.bed`) are passed to `smc++ vcf2smc` via the `--mask` option. This mask instructs SMC++ to treat TE regions as missing data, preventing invariant sites within TEs from being counted as non-coalescence observations (which would artificially inflate N_e estimates in TE-enriched windows).

2.4 Data preparation: `vcf2smc`

For each population, a **distinguished individual** is designated in each `smc++ vcf2smc` call (`-d IND IND` : same individual for both haplotypes, unphased diploid mode). The distinguished individual provides the "root" of the local genealogical tree at each position; coalescences between this individual and the rest of the population constitute the primary demographic signal.

To maximise sensitivity to recent N_e changes, **15 distinguished individuals** were used per population (or all individuals if $n < 15$), each producing one `.smc.gz` file per scaffold. Genotypes from *all* individuals in the population are included in each file.

Total:

- 15 distinguished individuals $\times$ 13 scaffolds = **195 .smc.gz files** per population
- 195 $\times$ 4 populations = **780 files** total, generated in parallel (GNU Parallel)

2.5 Cross-validated estimation: `smc++ cv`

$N_e(t)$ inference is performed with `smc++ cv` (k-fold cross-validation, $k = 3$), which automatically selects the regularisation parameter p . Three folds are used for p selection; a final model is then fitted on all data using the optimal p . This approach avoids both overparameterisation (too many breakpoints $\rightarrow$ artificial oscillations) and underparameterisation (trajectories too smooth). Output: `model.final.json`.

PARAMETER	VALUE	DESCRIPTION
--folds	3	3-fold cross-validation for p selection
--knots	32	Spline knots on the time axis
--em-iterations	30	EM algorithm iterations
μ (mutation rate)	5×10^{-9}	Sub./site/gen. — reference value
r (recombination rate)	3×10^{-8}	Per bp per gen. — angiosperm average; no genetic map available for <i>D. guianensis</i>
Generation time	30 yr	Conversion generations → calendar years
Distinguished individuals	15 / population	min(n, 15)
Super-Scaffolds used	13	Chromosome-level, ≥ 30 Mb

The four populations were processed sequentially (each `cv` run using 16 CPU cores). Ne(t) trajectories were exported as CSV via `smc++ plot -g 30 --csv`, converting the time axis from generations to calendar years. A mutation rate sensitivity analysis ($\mu \in \{2-8\} \times 10^{-9}$) was applied using the same axis rescaling as for Stairway Plot 2.

METHOD 3 — PAIRWISE DIVERGENCE

3. SMC++ split — divergence time between populations

3.1 Rationale

`smc++ split` jointly estimates the post-divergence Ne(t) trajectories of two populations and their divergence time `T_split`. The model extends the coalescent inference of each population by adding a merge event ("split") at `T_split` generations in the past. Initialisation is performed from the **marginal models obtained in the previous step** (`model.final.json` from SMC++ `cv`), ensuring faster and more robust convergence.

The analysis was performed for all **C(4,2) = 6 pairwise combinations** among the four clusters:

Pop_1/Pop_2 · Pop_1/Pop_3 · Pop_1/Pop_4 · Pop_2/Pop_3 · Pop_2/Pop_4 · Pop_3/Pop_4

3.2 Cross-population data preparation: vcf2smc

For divergence inference, `.smc.gz` files must contain genotypes from **both populations simultaneously**:

```
smc++ vcf2smc --mask te_merged.bed -d DIST DIST \
      VCF output.smc.gz SCAFFOLD \
      POP_A:samples_A POP_B:samples_B
```

For each pair, **10 distinguished individuals per population** were used (or all if $n < 10$):

- 10 dist. from Pop_A × 13 scaffolds = 130 files
- 10 dist. from Pop_B × 13 scaffolds = 130 files
- **260 .smc.gz files** per pair
- 6 pairs × 260 = **1,560 files** total (GNU Parallel)

3.3 Divergence inference: smc++ split

```
smc++ split --cores 16 -o pair_dir/estimate \
      model_A.final.json model_B.final.json \
      pair_dir/vcf2smc/*.smc.gz
```

The six pairs were processed sequentially. Output is a split model (`model.final.json`) containing Ne(t) trajectories for both populations and the `T_split` parameter.

3.4 Operational estimation of T_{split}

From Ne(t) trajectories exported as CSV, T_{split} is operationally estimated as the **oldest time point at which the relative Ne divergence between the two populations exceeds 5%**:

$$T_{split} = \max\{ t : |Ne_A(t) - Ne_B(t)| / \max(Ne_A(t), Ne_B(t)) > 0.05 \}$$

This criterion is applied on a common time grid by linear interpolation of both trajectories. It captures the point beyond which the two populations show distinct demographic histories, consistent with their effective separation. T_{split} values are reported in years before present (× generation time).

SYNTHESIS

4. Comparative parameter summary

PARAMETER	STAIRWAY PLOT 2	SMC++ CV	SMC++ SPLIT
Software (version)	v2.1.2	v1.15.5	v1.15.5
Input VCF	LD-pruned + TE-filtered	Not pruned, TE-filtered	Not pruned, TE-filtered
Demographic signal	SFS only	SFS + inter-variant distances	SFS + inter-variant distances
Reference μ	5×10^{-9}	5×10^{-9}	5×10^{-9}
r (recombination)	Not used	3×10^{-8}	3×10^{-8}
T _g (generation time)	30 yr	30 yr	30 yr
Uncertainty	95% CI (bootstrap)	Point estimate (cv)	Point estimate
TE masking	Via L (callable sites, GQ ≥ 20)	Via <code>--mask te_merged.bed</code>	Via <code>--mask te_merged.bed</code>
Distinguished individuals	N/A	15 / population	10 / pop. / pair
Super-Scaffolds used	All (global SFS)	13 chr-level ≥ 30 Mb	13 chr-level ≥ 30 Mb
Input files generated	1 SFS / population	195 .smc.gz / pop. (780 total)	260 .smc.gz / pair (1,560 total)
Temporal resolution	Best > 10,000 yr	Intermediate (100–500,000 yr)	Intermediate
μ sensitivity	✓ $\mu = 2-8 \times 10^{-9}$	✓ $\mu = 2-8 \times 10^{-9}$	—
Primary output	<code>.final.summary</code> (median Ne + CI)	<code>model.final.json</code> → CSV	<code>model.final.json</code> → CSV + T_split

FOR THE MANUSCRIPT

5. Writing notes for the M&M section

KEY REFERENCES

- **Stairway Plot 2:** Liu & Fu (2015) *Mol. Biol. Evol.* 32:2677–2693; Liu & Fu (2020) *Nat. Commun.*
- **SMC++:** Terhorst *et al.* (2017) *Nature Genetics* 49:303–309
- **easySFS:** Overcast *et al.* — auxiliary tool for SFS computation
- **bcftools:** Danecek *et al.* (2021) *GigaScience*

POINTS REQUIRING JUSTIFICATION IN THE MANUSCRIPT

- **$\mu = 5 \times 10^{-9}$:** document the source — is this estimated for *Dicorynia guianensis* specifically, or transferred from a closely related taxon (tropical Leguminosae)? Cite accordingly.
- **$r = 3 \times 10^{-8}$:** angiosperm average (e.g., Stapley *et al.* 2017) used as an approximation in the absence of a genetic map for the species. State this limitation explicitly.
- **$T_g = 30$ yr:** justify from species biology — age at sexual maturity, age structure of the population.
- **L for Pop_3** (193.7 Mbp vs. 215–221 Mbp for other clusters): the difference reflects reduced GQ coverage in this cluster, not a biological difference — mention in supplementary or as a note.
- **GONE / Stairway discordance for Pop_3:** the marked recent N_e reduction detected by GONE but not confirmed by Stairway Plot is directly addressed by the SMC++ vs. Stairway comparison (Fig. 3 of the SMC++ analysis). SMC++, with its intermediate temporal resolution, can help resolve the discordance between the two signals.
- **Outgroup exclusion:** state explicitly that the four Cameroon/Benin individuals (DBR_AGN, DBR_AGP, DBR_AGQ, DBR_AGR) were excluded from all demographic analyses.